

NLP Intelligence System: From Raw Text to Production

Build a fully operational sentiment classifier and search engine on two real-world noisy datasets designed, tracked, and deployed as a production-grade ML system using the MLOps practices from your previous course.

Assigned: Saturday, May 23

Due: Saturday, May 30 · 11:59 PM

Submission: GitHub repository

Weight: 10% of course grade

Overview

This lab bridges Lecture 1 theory with production engineering. You will apply every concept covered text normalization, tokenization, stemming, lemmatization, BoW, TF-IDF, BM25, and word embeddings on two deliberately contrasting datasets: [Amazon Fine Food Reviews](#) and [Sentiment140](#). The contrast is the lesson. The same pipeline, the same algorithms, the same code but the behavior will differ in ways that will force you to make real engineering decisions.

Unlike a standard notebook exercise, this lab is structured as a **software project**. You will version your data with DVC, track every experiment with MLflow, and ship your best model as a running FastAPI service inside a Docker container. By the end of the week, your GitHub repository should tell a clear end-to-end story: raw data in, live API out, with every decision along the way logged and reproducible.

Learning objectives

- **Build a reusable NLP pipeline:** Design a configurable preprocessing class that adapts to different text domains without rewriting code.
- **Compare vectorization methods:** Empirically demonstrate why TF-IDF and BM25 outperform raw counts and when they don't.

- **Understand distributional meaning:** Visualize word embeddings and articulate what classical vectors cannot capture.
- **Version data and experiments:** Use DVC for data versioning and MLflow to track all experimental runs reproducibly.
- **Build a working search engine:** Implement BM25-based retrieval with sentiment filtering over 500K+ real documents.
- **Deploy a production API:** Wrap classifier and search engine into a Dockerized FastAPI service with live endpoints.